

Multiple Choice (10 marks)

1. Compared to the variance of the Maximum Likelihood Estimate (MLE), the variance of the Maximum A Posteriori (MAP) estimate is _____
 - (a) higher
 - (b) same
 - (c) lower
 - (d) it could be any of the above
2. Which of the following is the joint probability of H, U, P, and W described by the given Bayesian Network $H \rightarrow U \leftarrow P \leftarrow W$? [note: as the product of the conditional probabilities]
 - (a) $P(H, U, P, W) = P(H) * P(W) * P(P) * P(U)$
 - (b) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(W | H, P)$
 - (c) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)$
 - (d) None of the above
3. Predicting the amount of rainfall in a region based on various cues is a _____ problem.
 - (a) Supervised learning
 - (b) Unsupervised learning
 - (c) Clustering
 - (d) None of the above
4. Statement 1| RELUs are not monotonic, but sigmoids are monotonic. Statement 2| Neural networks trained with gradient descent with high probability converge to the global optimum.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. Consider the Bayesian network given below. How many independent parameters would we need if we made no assumptions about independence or conditional independence $H \rightarrow U \leftarrow P \leftarrow W$?

- (a) 3
- (b) 4
- (c) 7
- (d) 15

True / False (10 marks)

Answer True or False

- 6. True or False: The rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ is 1.
- 7. True or False: Overfitting refers to a model that cannot generalize to new data but fits the training data perfectly.
- 8. True or False: $P(A, B, C)$ can be expressed as $P(A, B \mid C) * P(C)$ for Boolean random variables A, B, and C.
- 9. True or False: Neural networks can utilize a mix of different activation functions to improve learning and performance.
- 10. True or False: Lasso regression is more suitable for feature selection due to its ability to shrink some coefficients to zero.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Write a function to generate a two-dimensional array.
- 12. Write a function to find frequency count of list of lists.

End of Paper